Incremental LEFM Prediction of a Crack Trajectory from a Circular Hole

Abstract

Crack-direction prediction from a circular hole is examined as a recursively coupled numerical problem in quasi-static linear elastic fracture mechanics. The initiation point is selected from the maximum tensile tangential stress on the uncracked hole boundary. An initial segment follows its material-side normal, and subsequent directions are obtained from current-tip opening and sliding stress-intensity factors through the maximum tangential stress criterion. Each elastic solution retains the previously generated crack geometry. Reflection-paired tip meshes, prescribed-field extraction checks, independent crack-opening-displacement comparisons, and a centered-hole control support the numerical assessment. Fixed-geometry mesh comparisons and independently propagated mesh and increment studies distinguish extraction sensitivity from accumulated trajectory sensitivity. Mode mixity first increases and then changes sign as the crack develops, reversing the incremental turn while the opening intensity continues to increase. Characteristic path events persist under the tested mesh changes, while increment refinement contracts geometric and directional discrepancies and retains stable increment-normalized turning measures. These observations support numerical robustness within the tested families, rather than a rigorous error bound or a uniquely established boundary mechanism. The model addresses direction selection under prescribed increments; it does not resolve propagation kinetics, arrest, branching, or inelastic processes.

Keywords: crack trajectory; circular hole; linear elastic fracture mechanics; maximum tangential stress; mode mixity; interaction integral; crack-opening displacement.

1 Introduction

Circular holes are among the classical stress concentrators of fracture mechanics. Once a crack is prescribed at a hole boundary, its stress- intensity factors have been studied analytically for radial and inclined configurations since the early work of Bowie [1]; mixed-mode solutions for cracks emanating from circular holes were subsequently developed for finite plates [2], and modern weight-function formulations provide efficient mixed-mode SIF and crack-opening-displacement estimates for hole-edge cracks [3]. The onset problem is related but distinct. Finite-fracture-mechanics studies have examined crack initiation from an initially uncracked circular hole, including the competition between symmetric and asymmetric initiation [4] and the stability of subsequent growth under biaxial loading [5]. These results emphasize a basic distinction that is central here: identification of an initiation site does not by itself determine the trajectory of the finite crack that subsequently develops.

Once a finite crack exists, its direction depends on the mixed-mode field at its current tip and therefore on the complete crack geometry accumulated up to that state. The maximum tangential stress (MTS) criterion of Erdogan and Sih [6] provides a classical local direction rule in terms of the opening and sliding intensities K_I and K_{II}, respectively. Numerical crack-growth frameworks have long combined such local criteria with repeated updating or remeshing of the geometry; for example, Bouchard et al. [7] compared several mixed-mode direction criteria within an automatic finite-element propagation procedure. At a different modeling scale, cohesive-zone formulations replace the singular-tip idealization by a finite fracture process zone. Recent potential-based

cohesive calculations in a near-LEFM regime show that the selected kink direction can remain close to the classical elastic prediction even when the critical load is more sensitive to the cohesive-zone description [8]. The present work deliberately remains within LEFM in order to isolate the trajectory-selection problem itself.

Crack trajectories influenced by holes have been predicted using several different numerical and semi-analytical technologies. Wang et al. [9] treated curved cracks emanating from adjacent holes and compared predicted paths with fatigue experiments. Boundary-element propagation has been used for multiple cracks in perforated domains [10], while Fang et al. [11] combined adaptive extended isogeometric analysis with an interaction integral and the maximum circumferential-stress criterion for cracks interacting with arbitrary holes and voids. Their study also compared two prescribed crack-growth increments and found similar trajectories, with the smaller increment producing the smoother path. Yang et al. [12] used incremental remeshing to study cracks emanating from circular blasting holes and showed that, in mixed-mode configurations, the sign and magnitude of K_{II} govern the direction of curving. Related perforated-plate problems have been studied with a boundary-cracklet formulation [13], and commercial adaptive remeshing has been used to reproduce hole-induced crack deflection and arrest [14]. Taken together, these studies establish both the strong physical influence of holes on crack paths and the ability of several numerical technologies to reproduce such interactions.

For an incremental trajectory calculation, however, the stress-intensity factors are not merely postprocessed outputs. They determine the next segment, perturbing each later boundary-value problem. Numerical SIF extraction therefore enters the propagation algorithm recursively. The mixed-mode interaction-integral construction provides a natural energetic route to separating K_I and K_{II} [15]. Independent accuracy studies have shown that near-tip discretization and postprocessing choices can materially affect extracted SIFs: displacement-extrapolation estimates depend on element size, shape, and mesh arrangement [16]; energetic/domain-integral approaches are generally less dependent on detailed near-tip field fitting and are attractive for stepwise crack-growth calculations [17]; and comparative studies report their best finite-element SIF accuracy when domain integrals are combined with appropriately refined meshes [18]. Ghafoori Ahangar and Verreman [19] further demonstrated the relative robustness of the interaction-integral method with respect to angular discretization, element size, local refinement, crack length, and mixed-mode extraction. These results motivate treating verification of the SIF calculation as part of the crack-path problem rather than as an independent implementation detail.

The studies above provide extensive evidence for individual components of incremental LEFM propagation, but comparatively less attention has been given to the trajectory as a recursively generated numerical object whose characteristic events should remain stable under controlled perturbations of the numerical procedure. The purpose of the present study is therefore not to introduce a new mixed-mode fracture criterion or a new remeshing technology. Instead, we construct a single crack trajectory from a circular stress concentrator and ask which features of that trajectory can be regarded as robust consequences of the evolving elastic field. The uncracked-hole solution selects the initiation point and a material-side normal; only the first crack segment is prescribed along that normal. Every subsequent direction is obtained from the current-tip K_I and K_{II} values through the MTS rule after solving the complete current crack geometry.

The numerical assessment separates extraction accuracy from the sensitivity of a recursively generated trajectory. The interaction integral in equivalent domain form (EDI) is checked against prescribed asymptotic fields and independent crack-opening- displacement estimates. A centered-hole control tests whether the same EDI–MTS recurrence preserves and restores the exact straight symmetry path. Exterior-mesh density is perturbed independently, while a separate structured-tip resolution study varies the near-tip scale while holding the requested exterior sizing law fixed. Finally, crack-increment sensitivity is examined through independently generated 4-, 2-, and 1-mm paths using one dimensionless numerical family. Particular attention is paid to the

evolution of the mode mixity K_{II}/K_I, the location of its positive maximum and sign reversal, the corresponding change in turning sense, and the behavior of increment- normalized corrective quantities. This structure is intended to distinguish a physically interpretable trajectory from one that is only visually plausible within a particular discretization.

The contribution is a reproducible assessment of how initiation selection, current-tip extraction, and prescribed increments interact during recursive direction selection using established LEFM and MTS methods. Section 2 defines the geometry, loading, and stress-selected initiation frame. Section 3 develops the current-tip directional recurrence. Section 4 describes discretization, extraction, and verification, and Section 5 presents the trajectory and sensitivity results. Section 6 separates the mechanical interpretation from discretization evidence and model limitations. Section 7 summarizes the findings; the supplementary diagnostics document numerical acceptance.

2 Problem formulation

2.1 Geometry, material, and loading

Figure 1 introduces the plate geometry, the coordinate conventions, and the applied loading. The specimen has width $2A$ and height B, with a circular hole of radius R whose center is denoted by (x_c, y_c) in the schematic. The figure measures y_c upward from the lower plate edge. For the computational formulation we use a Cartesian frame with origin at the midpoint of the left edge, x pointing right and y upward. Thus

$$\Omega_{\mathrm{rect}} = [0, 2A] \times [-B/2, B/2], \qquad (x_c, y_c^{\mathrm{comp}}) = (x_c, y_c - B/2).$$

The curved line emerging from the hole illustrates an admissible crack path; it is schematic and does not prescribe the computed trajectory. The initiation point P_0 is selected from the uncracked-hole solution, and only the first crack direction is prescribed. Subsequent directions follow from the current-tip SIFs.

At step k, Ω_k is the material domain outside the hole and the retained crack Γ_k, with distinct upper and lower crack faces. The material is homogeneous, isotropic, and linearly elastic, with Young's modulus E and Poisson's ratio ν, under plane strain. For the crack geometry Γ_k ending at P_k, the displacement field $\boldsymbol{u}$, strain tensor $\boldsymbol{\varepsilon}$, and stress tensor $\boldsymbol{\sigma}$ satisfy

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0} \qquad \text{in } \Omega_k, \tag{1}$$

$$\boldsymbol{\varepsilon} = \tfrac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}}), \qquad \varepsilon_{33} = 0, \tag{2}$$

$$\boldsymbol{\sigma} = 2\mu\boldsymbol{\varepsilon} + \lambda\, \mathrm{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}, \quad \mu = \frac{E}{2(1+\nu)}, \quad \lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}. \tag{3}$$

Here $\boldsymbol{I}$ is the identity tensor, and μ and λ are the shear modulus and first Lamé coefficient, both with stress units. Uniform outward tensile tractions act on the horizontal edges, in the directions indicated by the arrows in Fig. 1. Their unsigned magnitude is $\sigma > 0$, as in the drawing. The traction vectors on the upper and lower edges are $(0, \sigma)$ and $(0, -\sigma)$, respectively. The vertical sides, hole boundary, and crack faces are traction free. Rigid-body motions are removed by imposing $u_x = u_y = 0$ at the lower-left corner and $u_y = 0$ at the lower-right corner. There are no body forces.

For fixed geometry, positive scaling of σ scales both SIFs equally, preserving their ratio and the MTS direction. The present formulation predicts the direction of imposed quasi-static increments, not a load-versus-time crack-growth history.

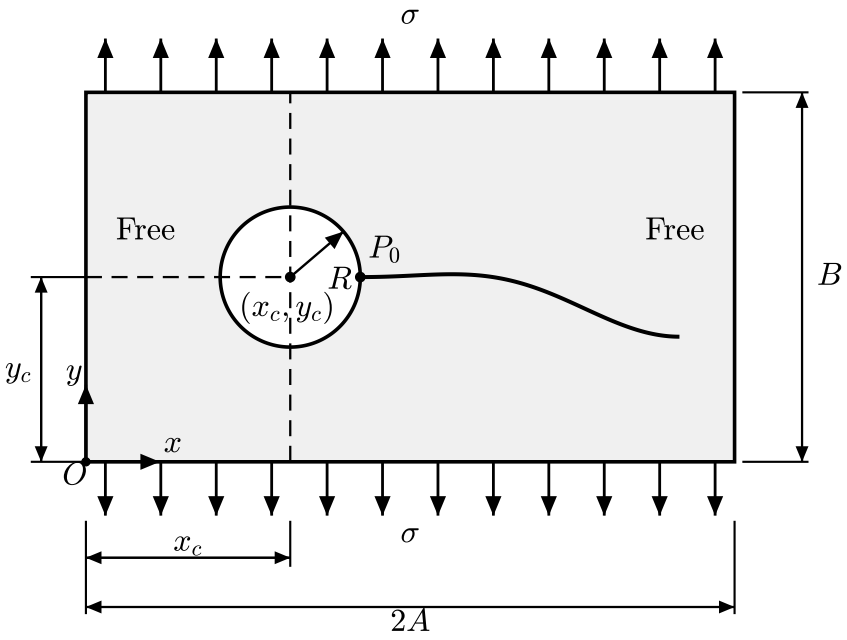


Figure 1: Specimen geometry and tensile loading. The plate has width $2A$ and height B, and the circular hole has radius R and center (x_c, y_c) in the schematic's lower-left coordinate system. The computational y coordinate is shifted by $-B/2$ relative to the sketch. The symbolic sketch is not drawn to the numerical scale. The arrows indicate outward tensile tractions of unsigned magnitude σ; the lateral edges, hole boundary, and crack faces are traction-free. The curvilinear crack is schematic, not a prescribed or independently solved trajectory.

2.2 Stress-selected initiation and the frozen frame

The uncracked-hole elastic field supplies the tangential boundary stress $\sigma_{tt}(\phi)$, where ϕ is measured counterclockwise from the positive Cartesian x direction. Its maximum tensile value selects the initiation point; the numerical boundary evaluation is described in Section 4. Let σ_c denote the tensile initiation threshold. The dimensionless initiation multiplier λ_{ini} is distinct from the Lamé coefficient λ. The normalized initiation rule is

$$\phi_* = \arg\max_{\phi} \sigma_{tt}^{+}(\phi;\sigma), \qquad \lambda_{\mathrm{ini}} = \frac{\sigma_c}{\max_{\phi} \sigma_{tt}^{+}(\phi;\sigma)}, \quad \sigma_{tt}^{+} = \max(\sigma_{tt}, 0). \tag{4}$$

The fitted angle ϕ_* specifies the initiation point and the outward material-side radial normal:

$$P_0 = \begin{bmatrix} x_c + R\cos\phi_* \\ y_c^{\mathrm{comp}} + R\sin\phi_* \end{bmatrix}. \tag{5}$$

The initiation condition supplies the location; it is not reused as a growth or arrest criterion along the crack.

The frozen basis is

$$\boldsymbol{n}_{\mathrm{mat}} = \begin{bmatrix} \cos\phi_* \\ \sin\phi_* \end{bmatrix}, \qquad \boldsymbol{t} = \begin{bmatrix} -n_{\mathrm{mat},y} \\ n_{\mathrm{mat},x} \end{bmatrix}. \tag{6}$$

The normal points radially out of the hole and into the material; $\boldsymbol{t}$ is its counterclockwise rotation. All absolute crack angles θ_k below are measured relative to this frozen normal, not to the global x axis. The global segment direction is therefore $\Theta_k = \phi_* + \theta_k$; translating the coordinate origin does not change this angle. With increment length Δa, the initial direction and first segment are prescribed by

$$\theta_1 = 0, \qquad P_1 = P_0 + \Delta a\, \boldsymbol{n}_{\mathrm{mat}}. \tag{7}$$

The initiation point and frame remain frozen for subsequent increments. The study treats one active crack; competing subsequent initiation sites are outside its scope.

3 Incremental crack-path formulation

3.1 Current-tip frame and stress-intensity factors

At P_k, the local forward direction is the *last* segment direction, $\boldsymbol{e}_1^{(k)} = (P_k - P_{k-1})/\Delta a$, with $\boldsymbol{e}_2^{(k)} = (-e_{1y}^{(k)}, e_{1x}^{(k)})$. This frame is not the mouth-to-tip chord. Local coordinates x_1, x_2 are measured from the current tip along these forward and transverse directions. With r measured ahead of the tip along x_1, the signed conventions correspond to

$$K_I^{(k)} = \lim_{r\to 0^+} \sqrt{2\pi r}\,\sigma_{22}(r,0), \qquad K_{II}^{(k)} = \lim_{r\to 0^+} \sqrt{2\pi r}\,\sigma_{12}(r,0). \tag{8}$$

Mode mixity is denoted by $q_K = K_{II}/K_I$. The signed intensities are evaluated by the interaction integral described below. The complete previous crack geometry is retained during each elastic solution.

3.2 The implemented MTS rule

Let α be a turn relative to $\boldsymbol{e}_1^{(k)}$, and write $c = \cos(\alpha/2)$ and $s = \sin(\alpha/2)$. The implemented angular coefficients are exactly

$$S(\alpha) = K_I^{(k)} c^3 - 3K_{II}^{(k)} s c^2, \tag{9}$$

$$T(\alpha) = K_I^{(k)} s c^2 + K_{II}^{(k)} c(1 - 3s^2). \tag{10}$$

Their physical stress amplitudes carry the common factor $1/\sqrt{2\pi r}$. Since $S' = -3T/2$, the direction rule finds a tensile stationary direction by solving $T(\alpha) = 0$. The tensile stationary root continuously connected to straight propagation is selected. All accepted states have positive opening-mode intensity and small turns. The implemented root selection and safeguards are documented with the software; they are unchanged in this study.

For small mode mixity, expansion of Eq. (10) gives $\alpha \simeq -2K_{II}/K_I$ in radians. This relation explains the turning sign but does not replace the root calculation used for the results.

3.3 Finite-increment recurrence

Define the direction vector in the frozen frame by

$$\boldsymbol{e}(\theta) = \cos\theta\, \boldsymbol{n}_{\mathrm{mat}} + \sin\theta\, \boldsymbol{t}. \tag{11}$$

The incremental and absolute angles are distinguished in the recurrence

$$\Delta\theta_{k+1} = \alpha_{\mathrm{MTS}}(K_I^{(k)}, K_{II}^{(k)}), \tag{12}$$

$$\theta_{k+1} = \theta_k + \Delta\theta_{k+1}, \tag{13}$$

$$P_{k+1} = P_k + \Delta a\, \boldsymbol{e}(\theta_{k+1}). \tag{14}$$

The accumulated crack length is $a_k = k\Delta a$; plot abscissae denoted by a refer to this length, not to the horizontal tip coordinate. Thus each fixed current geometry is qualified, solved once, and used to append one segment. Only the first direction and the increment length are prescribed. The accepted first-tip state seeds the subsequent recursion; its numerical values and the computed trajectory are given in the numerical study below.

4 Numerical implementation and verification

4.1 Numerical parameters and initiation state

For the computations, $2A = 300$ mm, $B = 200$ mm, and $R = 30$ mm. The center is $(x_c, y_c) = (170, 80)$ mm in the lower-left convention of Fig. 1; in the midheight computational frame it is $(170, -20)$ mm. The material constants are $E = 210$ GPa and $\nu = 0.30$ (plane strain). The reference tensile traction is $\sigma = 1$ MPa, and the tensile initiation threshold is $\sigma_c = 300$ MPa.

The numerical hole boundary uses a 480-segment polygon retaining the fitted initiation point. Stresses from six-node quadratic triangles (T6) at three material-side offsets are extrapolated linearly to the boundary, followed by a local quadratic angular fit. The accepted Stage-I solution yields $\phi_* = -1.5606^\circ$ and $P_0 = (0.199989, -0.0208170)$ m. The fitted peak stress under unit traction is 3.7141 MPa, so that $\lambda_{\text{ini}} = 80.774$. The frozen radial normal has components $(0.999629, -0.0272345)$. In the reference path, $\Delta a = 4$ mm; the separately computed increment-refinement families use 2 and 1 mm. All subsequent SIFs reported at reference traction have units $\text{MPa}\sqrt{\text{m}}$.

4.2 Geometry and paired tip discretization

The crack is represented by its complete polyline with distinct upper and lower face topology, including previous kink vertices. A temporary appended-hole carrier supports geometric identification; its crack faces are collapsed to the zero-width polyline while preserving separate face identities. A structured core with characteristic tip spacing h_{tip} is aligned with the last segment and paired under local reflection across that segment. The asymmetric physical exterior is retained through conforming graded triangulation. Three-node triangles (T3) are converted to straight-sided quadratic T6 elements.

The fixed production scales are

$$\frac{h_{\text{tip}}}{\Delta a} = 0.00675308135, \quad \frac{r_i}{\Delta a} = 0.10, \quad \frac{r_o}{\Delta a} = 0.65, \quad \frac{r_c}{\Delta a} = 0.75. \tag{15}$$

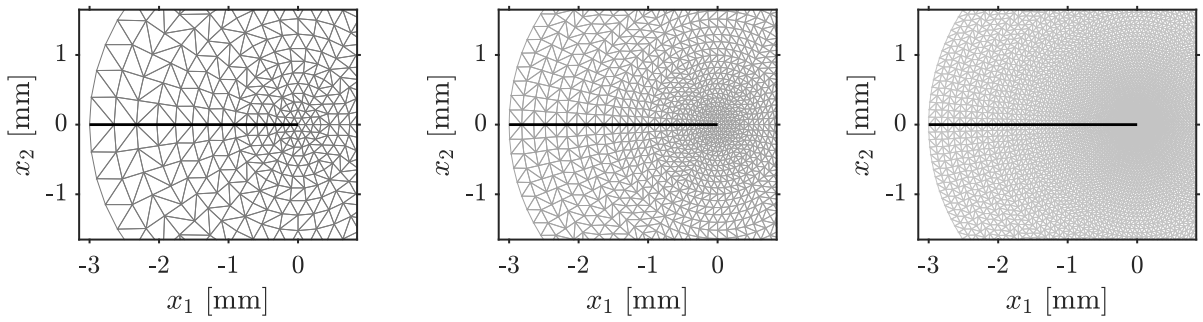
For this 4-mm increment these give $h_{\text{tip}} = 0.027012$ mm, $r_i = 0.4$ mm, $r_o = 2.6$ mm, and $r_c = 3$ mm. The grading transition is 4 mm and the far-field size cap is 2.5 mm. The paired core contains 12678 T3 triangles. All primary FE-nodal integration support lies inside this core, with 11316 participating elements.

Geometry and mesh checks retain the physical boundary vertices, distinct crack faces, shared intact ligament, conforming seams, positive element areas and T6 Jacobians. The mesh-angle and adjacent longest-edge-ratio gates are 20° and 1.8, respectively. These checks establish discretization integrity, not mesh-independent physical predictions.

For an exterior-mesh sensitivity test, a second mesh family, denoted M1, retains the same paired tip core, h_{tip}, r_i, r_o, and r_c, together with the same 4-mm transition length and all qualification gates. Only the exterior size law is relaxed: the far-field size cap is increased from 2.5 to 5.0 mm, the far-field grading slope from 0.10 to 0.15, and the boundary-metric growth parameter from 0.25 to 0.35. On the fixed P_{17} and P_{22} geometries this reduces the total T3 count by about 31–33% and the exterior-element count by about 40–43%, while leaving the 12678-element paired core and the 11316-element interaction- domain support unchanged. These fixed-geometry checks were used only to qualify M1 for a subsequent independent trajectory calculation.

4.3 Structured-tip resolution sensitivity

A separate fixed-geometry study varies the structured near-tip scale while keeping the accepted crack polyline, material, loading, interaction-domain radii, far-field cap, transition length, and



(a) $4h_0$ (b) $2h_0$ (c) h_0

Figure 2: Structured near-tip mesh hierarchy at a common core radius $r_c = 3$ mm and identical spatial limits; $h_0 = 0.027012$ mm is the baseline production tip size. The $4h_0$ mesh is shown for geometric illustration only: unlike $2h_0$ and h_0, it does not meet the native COD-sampling requirement.

MTS formulation unchanged. The production tip size is denoted by $h_0 = 0.027012$ mm. The three physically compared resolutions are

$$2h_0 = 0.054025 \text{ mm}, \qquad h_0 = 0.027012 \text{ mm}, \qquad h_0/2 = 0.013506 \text{ mm}. \tag{16}$$

These correspond to structured-core scale factors 2, 1, and 1/2. The paired-core radius remains $r_c = 3$ mm and the EDI annulus remains $r_i = 0.4$ mm to $r_o = 2.6$ mm. The paired cores contain 3318, 12678, and 49518 T3 elements, respectively. The established four COD windows contain 19/28/23/18 native points at $2h_0$, 38/55/44/34 at h_0, and 74/108/86/67 at $h_0/2$.

The dimensionless multiplier $\eta_c = h_{\text{tip}}/h_0$ controls structured-core resolution, while η_e multiplies the requested reference exterior sizing law. This comparison uses $\eta_c = 2$, 1, and 1/2 with $\eta_e = 1$. The exterior is remeshed to conform to each core interface, so its connectivity need not be identical across resolutions. The physical boundary, EDI radii, far-field cap, and transition length remain fixed. The independent M1 study supplies the complementary exterior-only perturbation with the paired tip core held fixed.

To test accumulation of this structured-tip resolution change, the $2h_0$ level is also propagated independently through the accepted 92-mm path. The prescribed first segment P_0P_1 is common, but the P_1 elastic problem is recomputed with the $2h_0$ family; its own $K_I^{(1)}$ and $K_{II}^{(1)}$ determine θ_2, and all later vertices follow recursively from the preceding $2h_0$ solution. No reference-path vertex after P_1 is imposed. The $h_0/2$ level is retained as fixed-geometry refinement evidence and is not propagated independently.

For geometric illustration, Fig. 2 also includes a coarser $4h_0$ core with $h_{\text{tip}} = 0.10805$ mm. It contains only 10/15/12/9 native COD points in the four established windows, so it fails the requirement of at least 12 points in every window. Consequently, $4h_0$ is shown only as part of the mesh-resolution hierarchy and is not used for physical SIF or MTS comparisons. No sampling or solver acceptance criterion is relaxed to admit it.

4.4 Interaction/domain integral

Let q be a radial domain weight and let m identify an auxiliary opening or sliding mode. Indices i, j refer to the two tip-local coordinates and repeated indices are summed; δ_{1j} is the Kronecker delta. For the actual field $(\boldsymbol{\sigma}, \boldsymbol{\varepsilon}, \boldsymbol{u})$, the interaction integral is

$$I^{(m)} = \int_{\Omega_k} \left[-(\boldsymbol{\sigma}^{(m)} : \boldsymbol{\varepsilon})\delta_{1j} + \sigma_{ij} u_{i,1}^{(m)} + \sigma_{ij}^{(m)} u_{i,1} \right] q_{,j} \, \mathrm{d}\Omega, \qquad i, j \in \{1, 2\}. \tag{17}$$

Actual gradients use the T6 displacement field. Analytical near-tip auxiliary stresses and centered finite-difference derivatives of auxiliary displacements are used, with the recorded derivative step

$h = \max(10^{-7}\ \mathrm{m}, 10^{-5}r)$. A radial nodal weight is one for $r \le r_i$, decreases linearly to zero at r_o, and is zero outside. Its T6 interpolant supplies the FE-consistent gradient in Eq. (17). Integration uses the 16-point triangular rule. For auxiliary amplitude $K_{\mathrm{aux}} = 1$ and plane strain,

$$K_m = \frac{E'}{2K_{\mathrm{aux}}} I^{(m)}, \qquad E' = \frac{E}{1-\nu^2}, \qquad m \in \{I, II\}. \tag{18}$$

Here $\mathrm{d}\Omega$ is the area measure, a comma denotes differentiation, and the colon denotes tensor contraction. The domain weight q is distinct from the mode-mixity ratio q_K. The normalization preserves the signed local-mode convention.

4.5 Independent COD postprocessing

Native upper/lower T6 face nodes are paired in the last-segment frame. If $[\![u_i]\!]$ denotes the upper-minus-lower displacement jump behind the tip at distance r, the apparent COD SIFs are

$$\begin{bmatrix} K_I^{\mathrm{app}}(r) \\ K_{II}^{\mathrm{app}}(r) \end{bmatrix} = \frac{\mu}{\kappa+1}\sqrt{\frac{2\pi}{r}} \begin{bmatrix} [\![u_2]\!] \\ [\![u_1]\!] \end{bmatrix}, \qquad \kappa = 3 - 4\nu. \tag{19}$$

Linear and quadratic fits in $r/\Delta a$ extrapolate these quantities to $r = 0$. The four fitting windows are $[0.04, 0.20]$, $[0.04, 0.30]$, $[0.08, 0.30]$, and $[0.12, 0.30]$ times Δa, with 38, 55, 44, and 34 native samples. COD verifies a second extraction route from the *same* elastic field. The production direction always uses the EDI SIFs; window/order scatter is not a statistical uncertainty interval.

4.6 Solver and evidence checks

The unclamped symmetric stiffness is reduced to free degrees of freedom, symmetrized within the audited formulation, and reordered by a symmetric approximate minimum-degree permutation. Symmetric Gauss–Seidel preconditioning is used with the preconditioned conjugate-gradient method (PCG). The reported-residual tolerance is 10^{-10}, the iteration limit is 5000, and the independently computed true-relative-residual gate is 5×10^{-10}. Constrained degrees of freedom remain exactly zero. For the finer fixed-geometry sensitivity meshes only, MATLAB PCG can report finite-precision stagnation immediately above the stated tolerance. In that case one Krylov restart from the returned iterate is permitted, using the same matrix, right-hand side, SGS factors, tolerance, and iteration limit. The final PCG and independently recomputed residual gates are unchanged; a restart therefore resets the recurrence but does not relax acceptance. Only an accepted elastic field is checkpointed, before SIF/COD/MTS postprocessing.

Prescribed nodal pure-I, pure-II, and tiny-mixed $(K_I, K_{II}) = (1, 10^{-4})$ fields test modal recovery and leakage on each qualified mesh. Recovery and tiny-mixed relative-error gates are 2×10^{-4}; the pure-mode cross-leakage and superposition gates are 10^{-10}. On the reference meshes, the observed maximum tiny-mixed relative error is 3.86×10^{-8}. These controls qualify extraction capability and topology rather than the global physical finite-element error.

The reference history contains accepted physical states through P_{23}. Its continuation P_{24} satisfies geometric and prescribed-field checks but has no accepted physical solution. For the accepted P_2–P_{23} fields, the largest true residual is approximately 1.00×10^{-10}, with 2442–3011 PCG iterations. Figure S1 collects these acceptance diagnostics; the saved full-precision records support independent verification of the reported trajectory and extraction quantities.

4.7 Crack-increment refinement family

The finite-step recurrence is assessed with three independently propagated paths using

$$\Delta a \in \{4, 2, 1\}\ \mathrm{mm}.$$

All three use the same dimensionless coarse numerical family rather than the production h_0 family. The structured core uses scale factor 2, so that

$$\frac{h_{\text{tip}}}{\Delta a} = 2(0.00675308135), \qquad \frac{r_i}{\Delta a} = 0.10, \qquad \frac{r_o}{\Delta a} = 0.65, \qquad \frac{r_c}{\Delta a} = 0.75. \tag{20}$$

The exterior uses the M1 dimensionless controls

$$\frac{h_{\text{far}}}{\Delta a} = 1.25, \qquad \frac{L_{\text{tr}}}{\Delta a} = 1.00,$$

with far-field grading slope 0.15 and boundary-metric growth 0.35. Therefore halving Δa halves the absolute core, EDI, transition, and exterior length scales while retaining the same numerical family. This is a refinement study of the complete fixed-increment algorithm in dimensionless form; it is not an experiment in which Δa alone is changed while an absolute mesh is held fixed.

The accepted Stage-I initiation point and frozen frame are identical for all three levels. The reserved first-segment length is changed only for Stage II, P_1 is rebuilt and solved independently, and every later vertex is generated from that path's own SIF/MTS history. Each trajectory is continued to a common total crack length of 92 mm. Pairwise geometric and direction differences are evaluated only at exact common crack lengths, using the 4-mm grid as the common set; no path interpolation is used for those comparisons. The location of local symmetry is reported separately by linear interpolation between the two solved states that bracket $q_K = K_{II}/K_I = 0$ and is therefore used only as a sub-increment diagnostic.

4.8 Centered-hole symmetric control

A separate control problem tests the directional recurrence under exact reflection symmetry. The full plate dimensions and hole radius are unchanged, but the circular hole is centered at $(0.150, 0)$ m. The crack starts from the rightmost point of the hole,

$$P_0 = (0.180, 0) \text{ m}, \qquad P_1 = (0.184, 0) \text{ m},$$

with the same increment $\Delta a = 4$ mm. The straight configuration is therefore a nominally pure mode-I path. To test its local directional stability rather than merely verify $K_{II} = 0$ at one geometry, five two-segment probes are prescribed,

$$\theta_2 \in \{-0.10^\circ, -0.05^\circ, 0, +0.05^\circ, +0.10^\circ\},$$

and each probe is rebuilt and solved independently with the same paired-core, EDI, solver, and MTS machinery used for the asymmetric calculation. The resulting one-step map is

$$F(\theta_2) = \theta_3 = \theta_2 + \alpha_{\text{MTS}}(K_I(\theta_2), K_{II}(\theta_2))\,. \tag{21}$$

At the straight state,

$$K_I(0) = 0.42541 \text{ MPa}\sqrt{\text{m}}, \qquad \frac{K_{II}(0)}{K_I(0)} = 6.22 \times 10^{-9},$$

and the predicted next direction is $F(0) = -7.13 \times 10^{-7}$ degrees, numerically indistinguishable from the symmetry line at the scale relevant here. The perturbed probes give the map summarized in Table 1.

The recovered opening-mode intensity is even with respect to the imposed perturbation, whereas K_{II}/K_I and $F(\theta_2)$ are antisymmetric to within numerical mismatches of order 10^{-9}. The two symmetric finite-difference estimates are

$$\frac{F(+0.05^\circ) - F(-0.05^\circ)}{0.10^\circ} = -0.11416, \qquad \frac{F(+0.10^\circ) - F(-0.10^\circ)}{0.20^\circ} = -0.11415,$$

Table 1: One-step directional map for the centered-hole symmetric control. Each row is an independently rebuilt and solved two-segment geometry.

θ_2 [deg]	$F(\theta_2) = \theta_3$ [deg]
-0.10	$+0.011415$
-0.05	$+0.005708$
0	-7.13×10^{-7}
$+0.05$	-0.005708
$+0.10$	-0.011415

and a least-squares fit constrained through the origin gives

$$F'(0) = -0.11415. \tag{22}$$

Because $|F'(0)| < 1$, the straight symmetric path is a locally attracting fixed path for this one-step discrete recurrence. The negative multiplier means that a small angular perturbation changes sign after one update while its magnitude is reduced by almost an order of magnitude. This is a local stability statement for the chosen 4-mm recurrence, not a proof of global path stability. Crack-increment dependence is assessed independently by the three-level refinement family described above.

5 Crack-trajectory results

5.1 Accepted path and direction history

The accepted first-tip SIFs at the stated unit traction are $K_I^{(1)} = 0.36648$ and $K_{II}^{(1)} = 4.1983 \times 10^{-6}$ MPa$\sqrt{\text{m}}$, giving $\Delta\theta_2 = \theta_2 = -0.0013127°$. The subsequent accepted elastic states are P_2 through P_{23}.

Figure 3 distinguishes the accepted P_0–P_{23} polyline from the dashed, qualified but unsolved P_{23}–P_{24} extension. The accepted crack length is 92 mm. The last tip and its horizontal clearance from the right boundary are

$$P_{23} = (291.840, -25.579) \text{ mm}, \qquad 2A - x_{23} = 8.16 \text{ mm}. \tag{23}$$

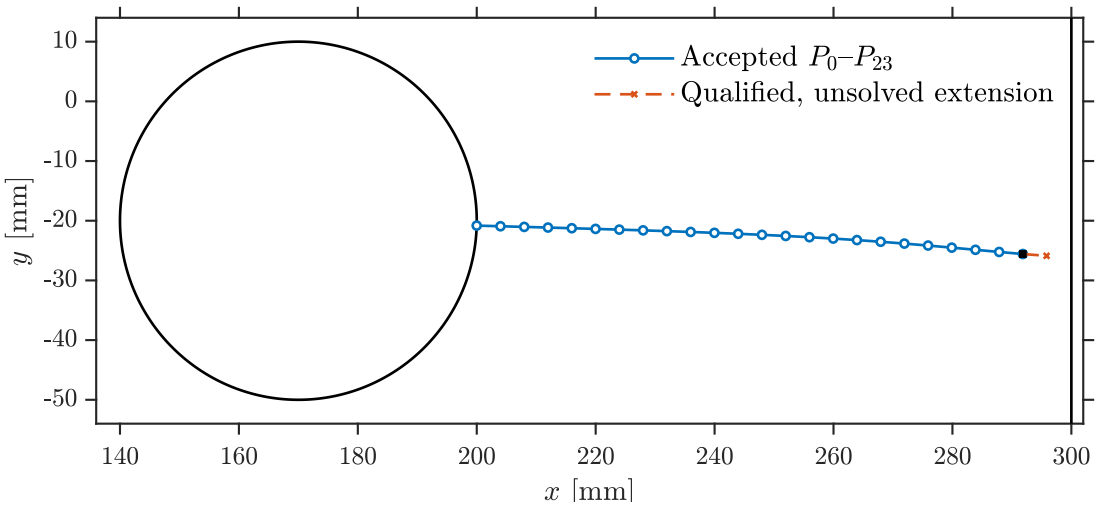
Directions vary gradually, although the finite-increment representation is piecewise linear rather than a continuously smooth curve.

The absolute direction becomes increasingly negative up to $\theta_{22} = -3.646°$, then rotates toward a less negative inclination at P_{23} (Fig. 4). A positive late turn does not imply an upward-moving tip: the accepted segments remain negatively inclined and their global y coordinates continue to decrease.

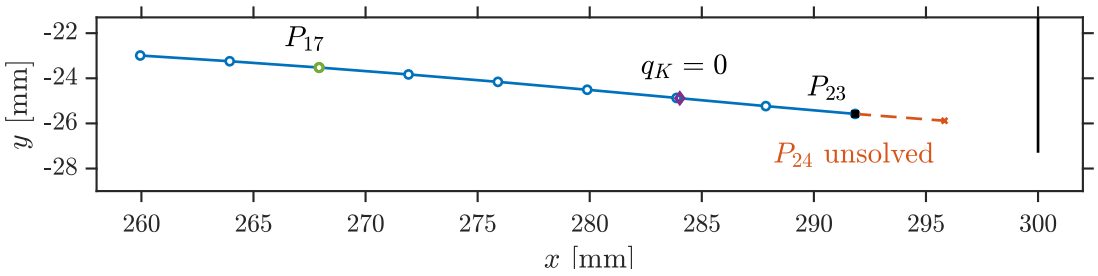
5.2 Mode intensities and the positive maximum

Figure 4 shows monotonically increasing K_I, from 0.36648 at P_1 to 1.2444 at P_{23}. The initially very small positive K_{II} grows to its positive maximum at P_{17}, where $a = 68$ mm, $K_{II} = 0.0025484$, and $q_K = K_{II}/K_I = 0.0032389$. The positive mode-mixity maximum occurs at the same state. It is not the maximum *absolute* mode-II intensity: the negative mode-II magnitude at the last accepted tip is larger.

From P_{17} through P_{21}, K_{II} remains positive but decreases. Accordingly the negative incremental MTS turn becomes smaller in magnitude (Fig. 5). This change in turning response accompanies an increasing opening-mode intensity, not unloading.

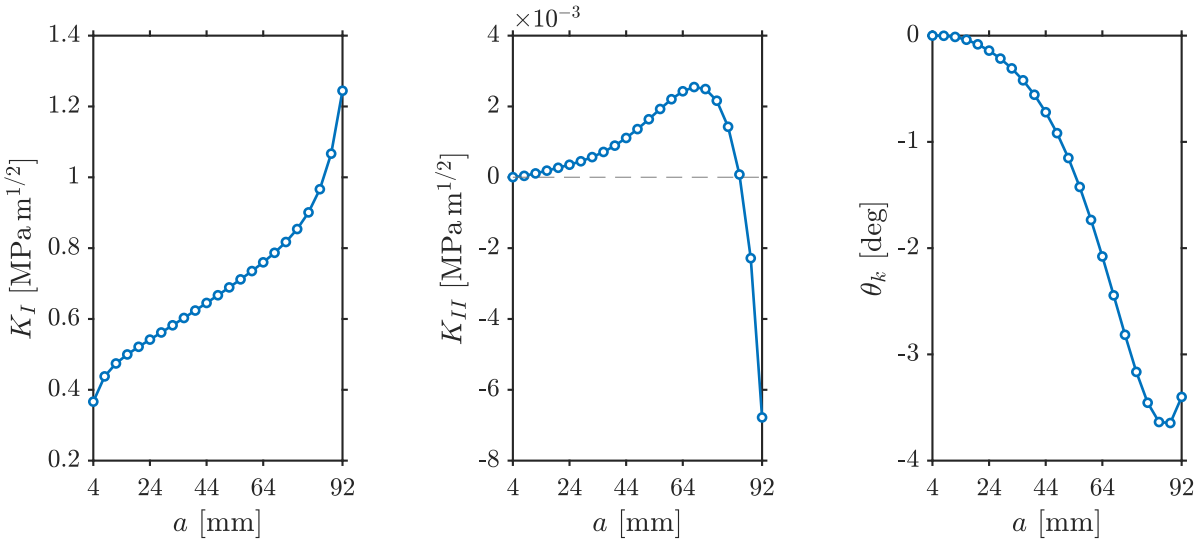


(a) Hole and accepted trajectory through P_{23}, with the qualified but physically unsolved P_{23}–P_{24} extension.



(b) Late-path detail with equal spatial scales. The markers identify P_{17}, the linearly interpolated $K_{II}/K_I = 0$ location, P_{23}, and the qualified unsolved point P_{24}.

Figure 3: Hole-initiated trajectory and late detail. Both spatial panels use equal x and y scales within the panel. The circle gives geometric context; the numerical hole is polygonally discretized. Solid segments are accepted geometry through P_{23}. The dashed extension is qualified geometry only and has no accepted physical SIFs. The diamond is a linear interpolation marker, not a solved tip.



(a) Opening-mode factor K_I. (b) Sliding-mode factor K_{II}. (c) Absolute direction θ_k.

Figure 4: Accepted mode-I and mode-II stress-intensity factors (MPa$\sqrt{\text{m}}$) and absolute direction in the frozen initiation frame, all plotted against the accumulated crack length a (mm). Intensities correspond to unit remote traction. The direction minimum occurs at P_{22}; a positive subsequent turn yields the less negative P_{23} direction. No physical curve includes P_{24}.

Table 2: Characteristic accepted states. SIFs have units MPa$\sqrt{\text{m}}$; angles are in degrees. The interpolated row contains only a graphical zero estimate, not a solved stress field or crack direction.

State	a [mm]	K_I	K_{II}	K_{II}/K_I	θ_k	$\Delta\theta_{k+1}$
P_1	4	0.36648	4.198×10^{-6}	1.146×10^{-5}	0.0000	-0.00131
P_{17}	68	0.78679	2.548×10^{-3}	3.239×10^{-3}	-2.4444	-0.37115
P_{21}	84	0.96631	7.860×10^{-5}	8.134×10^{-5}	-3.6366	-0.00932
P_{22}	88	1.0667	-2.284×10^{-3}	-2.142×10^{-3}	-3.6460	+0.24541
P_{23}	92	1.2444	-6.780×10^{-3}	-5.449×10^{-3}	-3.4006	+0.62435
Interpolated	84.15	—	—	0	—	—

5.3 Local-symmetry crossing and late turning reversal

At P_{21} ($a = 84$ mm), $q_K = +8.134267 \times 10^{-5}$; at P_{22} ($a = 88$ mm), $q_K = -0.0021416$. The sign change is therefore bracketed by two solved states. For visualization only, linear interpolation gives

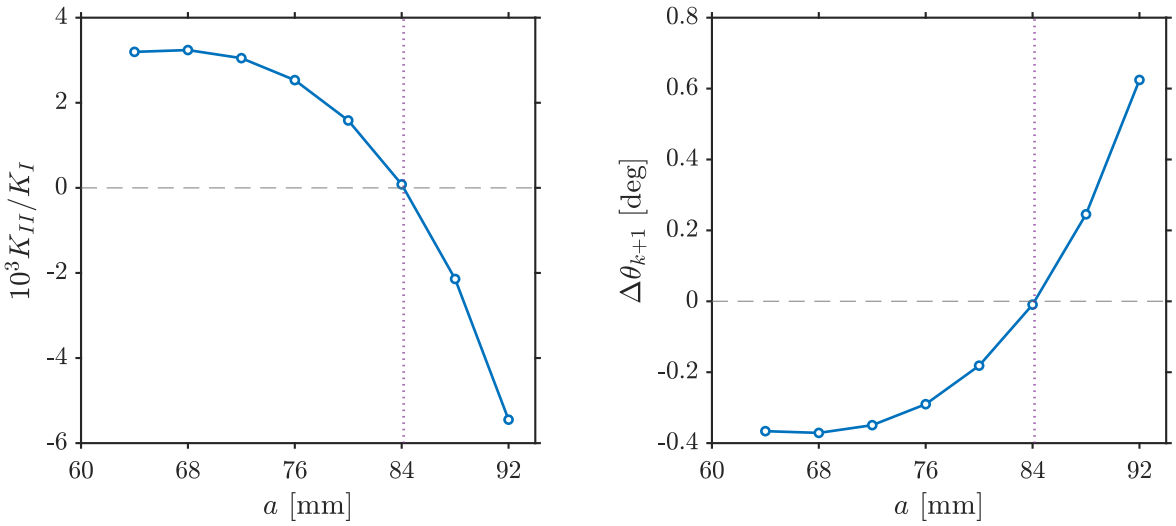
$$a_{\text{LS}} = a_{21} - \frac{q_K^{(21)}}{q_K^{(22)} - q_K^{(21)}}(a_{22} - a_{21}) \simeq 84.15 \text{ mm}. \tag{24}$$

This estimate is not an independently solved local-symmetry state and does not quantify interpolation or increment-size error.

Mode II is negative at P_{22} and P_{23}, so the next turning increments are positive. The last accepted state has $K_I = 1.2444$, $K_{II} = -0.0067803$, and $q_K = -0.0054488$. It predicts $\Delta\theta_{24} = 0.62435°$ and $\theta_{24} = -2.7762°$. The latter defines the next qualified geometry, not a physical solution at P_{24}.

5.4 COD comparison and extraction sensitivity

All eight stored COD fit definitions retain positive mode mixity at P_{21} and negative mode mixity at P_{22}, corroborating the bracketed sign change without making COD the direction selector. Differences remain nonzero and depend on the fitting definition (Fig. 6). Over all 176 fits, the



(a) Late mode-mixity history.

(b) Incremental MTS turn.

Figure 5: Mode-mixity sign reversal and opposite-signed MTS turning. Each marker is an accepted current-tip state; its turn predicts the next segment. The vertical dotted line is the interpolated $K_{II}/K_I = 0$ location. The positive turn reported at $a = 92$ mm belongs to the P_{23} solution and predicts the unsolved next geometry.

largest absolute COD–EDI turning difference is 0.01024°, at P_{23} for a linear fit over $[0.12, 0.30]\Delta a$. The quadratic-fit turns at P_{23} range from 0.6204° to 0.6225°, compared with the EDI value 0.62435°.

Near local symmetry, relative differences can be large despite small absolute errors. At P_{21} the same rear-window linear fit gives mode mixity about 26.3% above EDI, while its turning-angle difference is only 0.00245°. Thus visual overlap of full-history curves is not sufficient to establish extraction accuracy. The late fitting sensitivity is a postprocessing limitation to investigate, not an error bar on the physical path. Both routes also share any error in the underlying elastic solution.

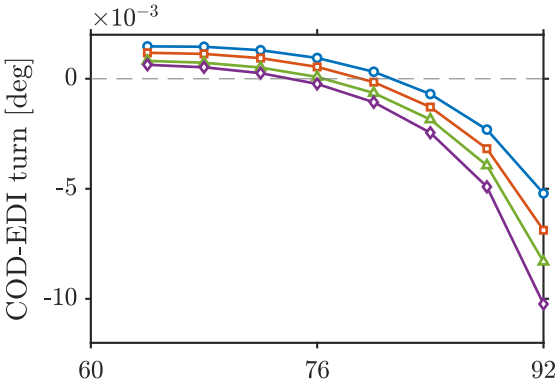
5.5 Sensitivity to structured-tip resolution

For comparisons at matched crack lengths, δx and δy denote coordinate differences between the compared and reference paths, $\delta r = (\delta x^2 + \delta y^2)^{1/2}$ is their separation, and $\delta\theta$ is the difference of absolute directions in the frozen frame. The symbol $\Delta\theta_{k+1}$ remains reserved for a local MTS turn. Differences in mode mixity and interpolated crossing length are denoted by δq_K and δa_{LS}, respectively.

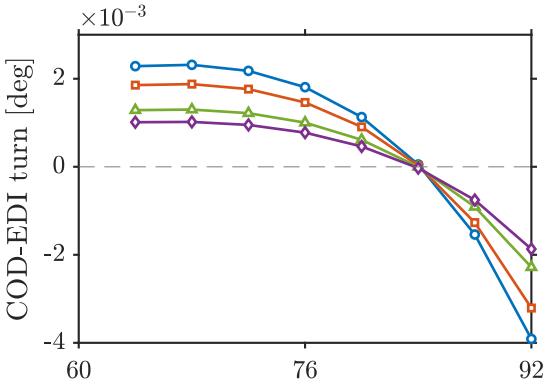
The fixed-geometry three-level comparison changes the structured-tip resolution with the requested exterior sizing law held at its reference value. At P_{17}, halving the production tip size from 0.027012 to 0.013506 mm changes K_I by 0.00134%, K_{II}/K_I by 1.05×10^{-6} in absolute value, and the MTS turn by only -1.20×10^{-4} degrees. Thus the positive mode-mixity maximum is stable under this structured-tip refinement at fixed requested exterior sizing.

The more discriminating comparison is the P_{21}–P_{22} bracket, where K_{II} changes sign. Table 3 shows three physically qualified levels spanning a factor of four in h_{tip}. Every level keeps $K_{II}/K_I > 0$ at P_{21} and $K_{II}/K_I < 0$ at P_{22}, so the local-symmetry bracket and the associated reversal of MTS turning sense are unchanged.

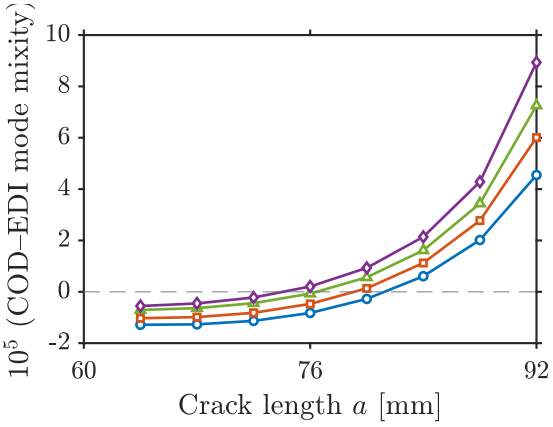
The interpolated crossing moves from 84.14027 mm at $2h_0$ to 84.14637 mm at h_0 and 84.14941 mm at $h_0/2$. The successive changes, 6.10 μm and 3.04 μm, are tiny relative to the 4-mm crack increment. Taken only as a three-level diagnostic, their ratio corresponds to an observed order



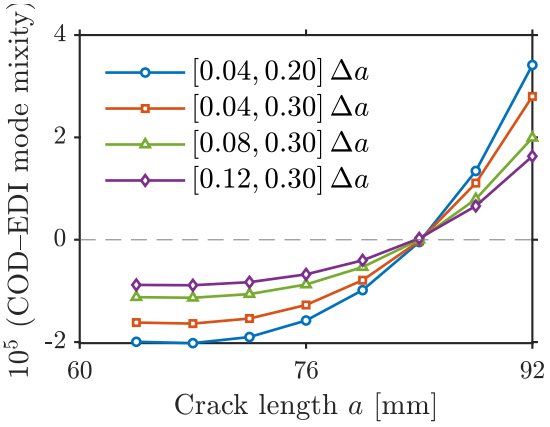
(a) Turning-angle differences for linear COD fits.



(b) Turning-angle differences for quadratic COD fits.



(c) Mode-mixity differences for linear COD fits.



(d) Mode-mixity differences for quadratic COD fits.

Figure 6: Late-path extraction differences for all four native COD windows and both polynomial degrees. Window bounds are multiples of Δa. Zero is equality with EDI, not a known exact solution. The upper panels show differences in the MTS turns computed from the two sets of SIFs; the lower panels show differences in mode mixity. The largest observed difference occurs at the last accepted tip; individual difference curves need not vary monotonically.

Table 3: Fixed-geometry sensitivity to structured-tip resolution. The crack geometries at P_{21} and P_{22}, material, loading, EDI radii, far-field cap, transition length, and MTS formulation are unchanged; the structured tip scale changes while $\eta_e = 1$ fixes the requested exterior sizing law. Exterior connectivity need not be identical. The final column is an interpolation between solved values, not a solved crack state; its extra digits resolve numerical mesh-comparison shifts only.

Resolution	h_{tip} [mm]	$q_K(P_{21})$	$\Delta\theta_{22}$ [deg]	$q_K(P_{22})$	$\Delta\theta_{23}$ [deg]	a_{LS} [mm]
$2h_0$	0.054025	$+7.7957 \times 10^{-5}$	−0.008933	-2.1451×10^{-3}	+0.245805	84.14027
h_0	0.027012	$+8.1343 \times 10^{-5}$	−0.009321	-2.1416×10^{-3}	+0.245406	84.14637
$h_0/2$	0.013506	$+8.3031 \times 10^{-5}$	−0.009515	-2.1399×10^{-3}	+0.245207	84.14941

of approximately 1.00 for this interpolated quantity. The exterior is remeshed to conform to each core and the crossing is inferred between states separated by one full crack increment. This three-level regularity is therefore not treated as a formal convergence order or as micrometer-scale physical accuracy.

The independent $2h_0$ trajectory tests whether these small fixed-geometry differences accumulate along the path. Figure 7 uses the same four-panel comparison as the exterior-mesh study. The trajectories are visually indistinguishable at specimen scale [Fig. 7a]. Over the full accepted history, the maximum vertical-coordinate difference is 0.12760 μm and the maximum Euclidean separation of corresponding tips is 0.12800 μm, both at P_{23}. The maximum absolute direction difference is 0.24539 millidegrees at P_{22} [Figs. 7b and 7c].

The mechanical history is equally stable. Both mesh families place the positive maximum of K_{II}/K_I at P_{17} and retain positive mode mixity at P_{21} and negative mode mixity at P_{22} [Fig. 7d]. The maximum absolute difference in K_{II}/K_I is 2.65×10^{-7}. Linear interpolation of the two independently propagated P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for h_0 and 84.146276 mm for $2h_0$, a difference of -0.000094 mm (-0.094 μm). As above, this is only a sub-increment sensitivity diagnostic, not a micrometer-accurate physical location of local symmetry.

The fixed-geometry and independent-path results therefore show that the reported characteristic events are insensitive to the tested structured-tip resolution changes with the requested exterior sizing law fixed. This parameter isolation does not imply identical exterior connectivity or bound every discretization error.

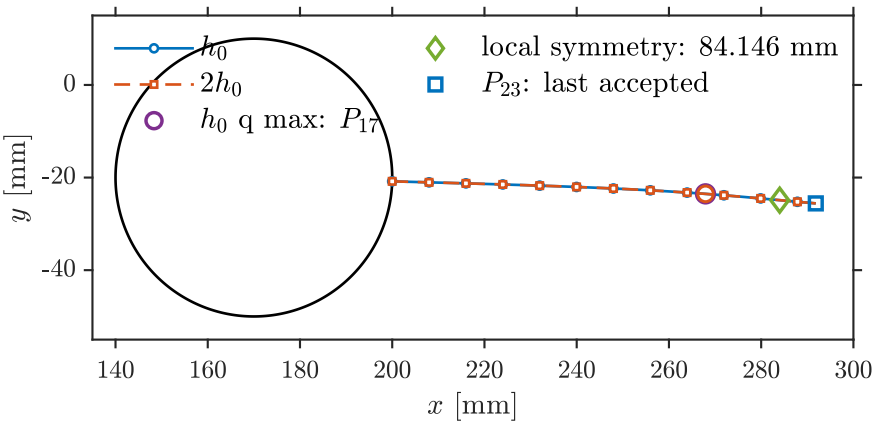
5.6 Sensitivity to exterior-mesh density

The fixed-geometry M1 checks establish that the coarser exterior mesh does not materially perturb the extracted current-tip quantities when the crack polyline is held fixed. To test accumulated trajectory sensitivity, however, the M1 path was regenerated independently rather than evaluated on the reference vertices. The prescribed first segment P_0P_1 is common to both calculations, but the P_1 elastic problem is resolved on M1, its own $K_I^{(1)}$ and $K_{II}^{(1)}$ determine θ_2, and every subsequent vertex follows recursively from the preceding M1 solution. Thus no reference-path vertex after P_1 is imposed.

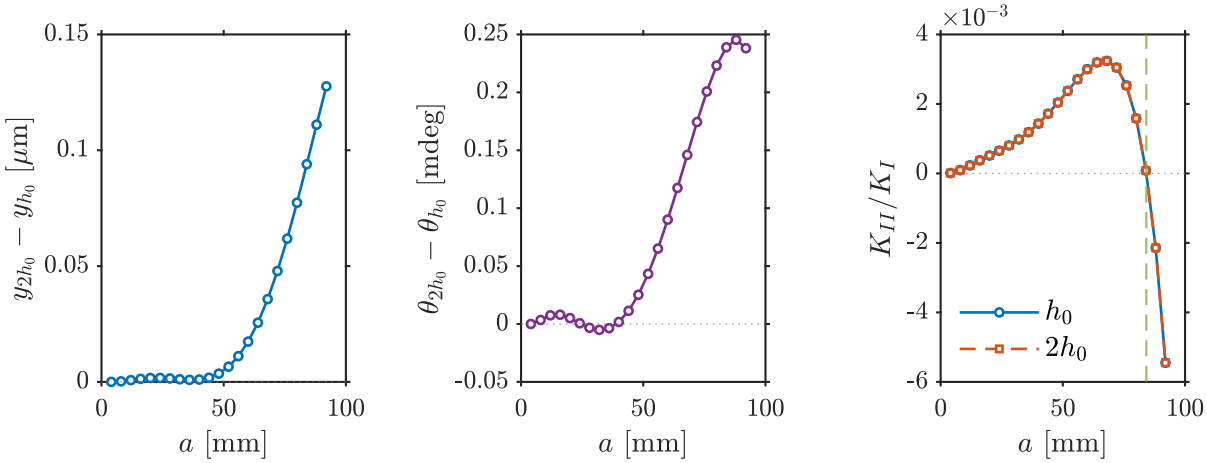
Figure 8 compares the two complete accepted histories through P_{23}. At specimen scale the trajectories are visually indistinguishable [Fig. 8a]. The accumulated difference is resolved by plotting the coordinate and direction differences directly. Over all common accepted states, the maximum vertical-coordinate difference is 0.06734 μm, the maximum Euclidean separation of corresponding tips is 0.06742 μm, and the maximum absolute direction difference is 0.07304 millidegrees, or 7.30×10^{-5} degrees [Figs. 8b and 8c].

The mechanical history is equally stable. Both calculations place the positive maximum of K_{II}/K_I at P_{17} ($a = 68$ mm), and both retain positive mode mixity at P_{21} and negative mode mixity at P_{22} [Fig. 8d]. The maximum absolute difference in K_{II}/K_I over the accepted histories is 2.89×10^{-7}. Linear interpolation of the two respective P_{21}–P_{22} brackets gives $a_{\mathrm{LS}} = 84.14637$ mm for the reference trajectory and 84.146 mm for M1, a difference of 0.000322 mm (0.322 μm). This sub-increment difference is reported only as a sensitivity diagnostic; neither interpolated value has physical precision at the micrometer scale.

The result therefore supports robustness of the reported mode-mixity maximum, sign reversal, and late turning response with respect to a substantial coarsening of the *exterior* mesh. It is deliberately narrower than a general mesh-independence claim because the paired tip core and interaction-domain radii are held fixed. Local tip-resolution sensitivity is examined separately above, whereas interaction-domain-radius sensitivity remains unresolved.



(a) Independently propagated trajectories for the production h_0 mesh family and the coarser $2h_0$ structured core at fixed requested exterior sizing. The two paths are visually indistinguishable at the scale of the specimen.

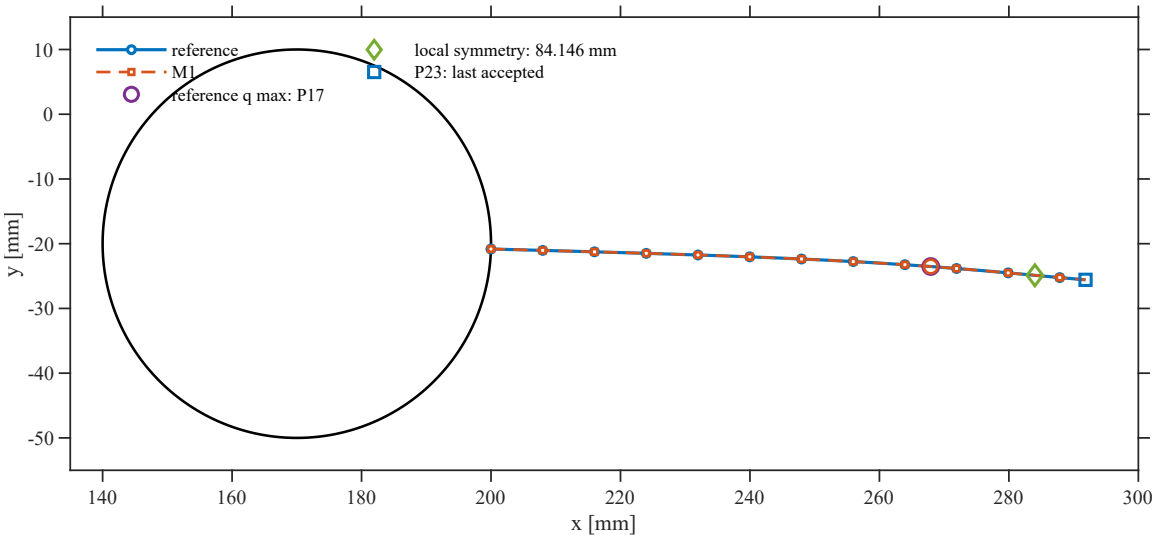


(b) Accumulated vertical tip-coordinate difference, $y_{2h_0} - y_{h_0}$.

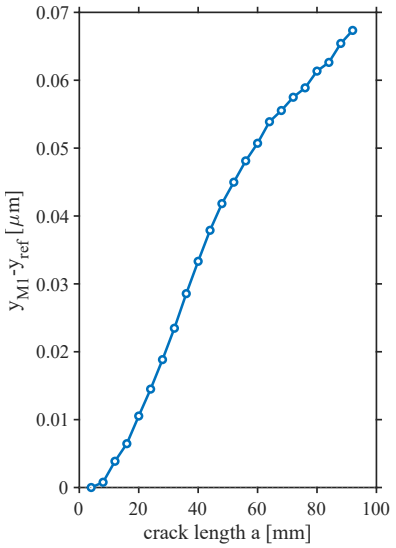
(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{2h_0} - \theta_{h_0}$.

(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

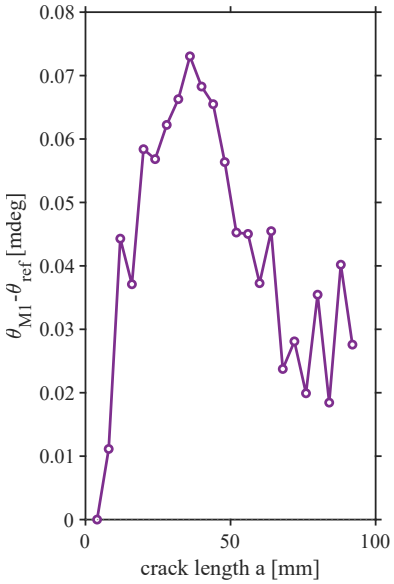
Figure 7: Sensitivity of the incremental LEFM trajectory to structured-tip resolution. The $2h_0$ calculation doubles the production tip size from $h_0 = 0.027012$ mm to 0.054025 mm while retaining the interaction-domain radii, far-field cap, and transition length. $\eta_e = 1$ preserves the requested exterior sizing law; the conforming exterior connectivity need not be identical. The $2h_0$ trajectory is propagated independently from a recomputed P_1 field rather than from reference vertices. The local-symmetry marker is the linearly interpolated reference crossing, not an independently solved state.



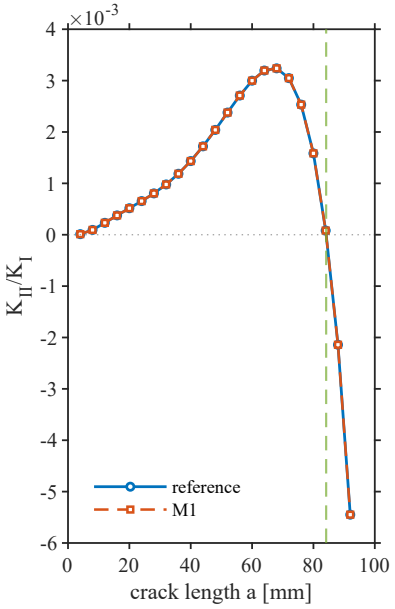
(a) Actual crack trajectories obtained with the reference exterior mesh and the independently propagated M1 exterior mesh. The two paths are visually indistinguishable at the scale of the specimen.



(b) Accumulated vertical tip-coordinate difference, $y_{\mathrm{M1}} - y_{\mathrm{ref}}$.



(c) Difference in the absolute crack direction in the frozen initiation frame, $\theta_{\mathrm{M1}} - \theta_{\mathrm{ref}}$.



(d) Mode-mixity ratio K_{II}/K_I along the two independently propagated trajectories.

Figure 8: Sensitivity of the incremental LEFM trajectory to coarsening of the exterior mesh. The M1 calculation retains the audited crack-tip core and EDI extraction region but uses a substantially coarser exterior mesh. The comparison is based on two independently propagated trajectories: the M1 calculation recomputes the P_1 SIFs and every subsequent MTS update rather than reusing the reference vertices. The local-symmetry diamond marks the interpolated reference crossing, not a solved tip.

Table 4: Successive differences between independently propagated paths in the dimensionless crack-increment refinement family. Comparisons use exact common crack lengths; no path interpolation is used.

Comparison	$\max \lvert\delta y\rvert$ [μm]	$\max \delta r$ [μm]	$\max \lvert\delta\theta\rvert$ [mdeg]	$\max \lvert\delta q_K\rvert$
2 mm − 4 mm	35.4181	35.4721	214.931	2.48578×10^{-3}
1 mm − 2 mm	18.2629	18.2911	127.391	1.41618×10^{-3}

Table 5: Characteristic quantities in the crack-increment refinement family. The local-symmetry location is obtained by linear interpolation between the two solved states bracketing $q_K = 0$.

Δa [mm]	$a(q_{K,\max})$ [mm]	$q_{K,\max}$	$q_{K,\max}/\Delta a$ [mm^{-1}]	$(\Delta\theta/\Delta a)_{q_{\max}}$ [deg/mm]	a_{LS} [mm]
4	68	3.2394×10^{-3}	8.0984×10^{-4}	−0.092799	84.148913
2	66	1.6127×10^{-3}	8.0633×10^{-4}	−0.092398	83.550120
1	66	8.0407×10^{-4}	8.0407×10^{-4}	−0.092139	83.268112

5.7 Crack-increment sensitivity

The three-level increment study compares independently generated $\Delta a = 4$, 2, and 1 mm trajectories over the exact common range 4–92 mm. Figure 9 shows the three actual paths together with the successive vertical and directional differences and the turning density. At specimen scale the trajectories nearly overlap [Fig. 9a], whereas the difference panels resolve a systematic reduction of the accumulated path error when the increment is halved [Figs. 9b and 9c].

Table 4 quantifies these successive differences. Halving the increment from 4 to 2 mm gives a maximum corresponding-tip separation of 35.472 μm; halving it again to 1 mm reduces that maximum to 18.291 μm. The corresponding maximum vertical-coordinate differences are 35.418 and 18.263 μm. The maximum absolute direction difference likewise falls from 214.931 to 127.391 millidegrees.

The successive contraction ratios are 1.939 for the maximum tip separation and 1.687 for the maximum direction difference. If these ratios are converted only as diagnostics through $p = \log_2(d_{\Delta a}/d_{\Delta a/2})$, where d denotes the selected successive-difference maximum, they correspond to $p \simeq 0.96$ and $p \simeq 0.75$, respectively. Because the trajectory is generated recursively and the reported quantities are maxima that need not occur at the same crack length, these values are not treated as formal Richardson orders. They do, however, show clear contraction of both the accumulated path and direction differences under successive halving of the increment.

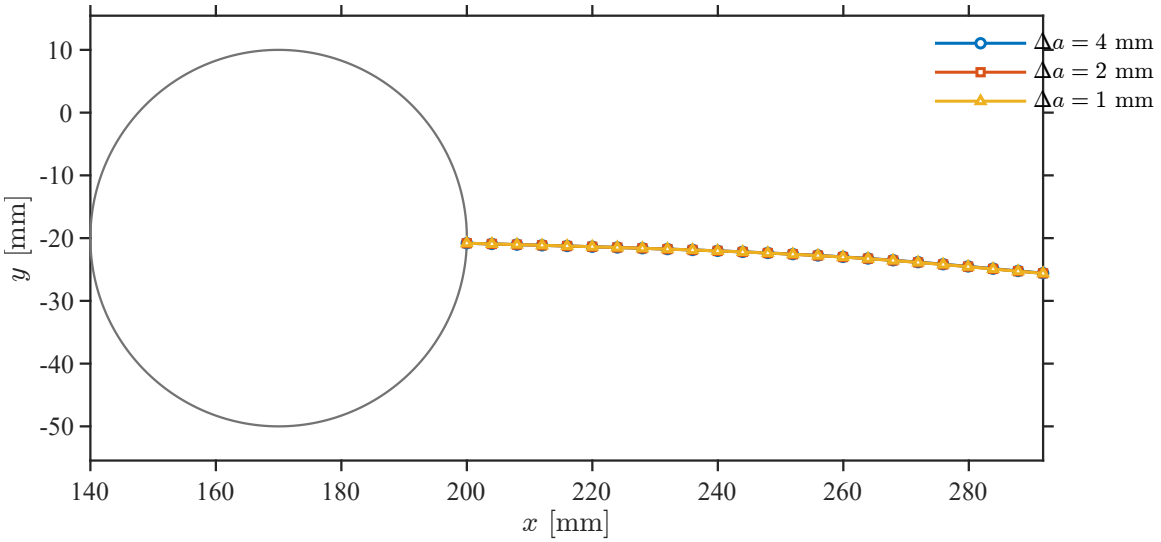
A more revealing pattern appears in the mode-mixity correction. Table 5 lists the positive maximum of $q_K = K_{II}/K_I$ for each level. The raw maximum decreases almost in direct proportion to Δa: from 3.23937×10^{-3} at 4 mm to 1.61266×10^{-3} at 2 mm and 8.04068×10^{-4} at 1 mm. In contrast, the normalized quantity $q_{K,\max}/\Delta a$ changes by only about 0.71% over the entire 4-to-1-mm refinement. The MTS turning density at the same characteristic state is similarly stable, varying from −0.092799 to −0.092139 deg/mm. The complete turning-density histories in Fig. 9d show that this agreement is not confined to a single state: refinement primarily resolves the same smooth variation on progressively finer crack-length grids.

This behavior is consistent with

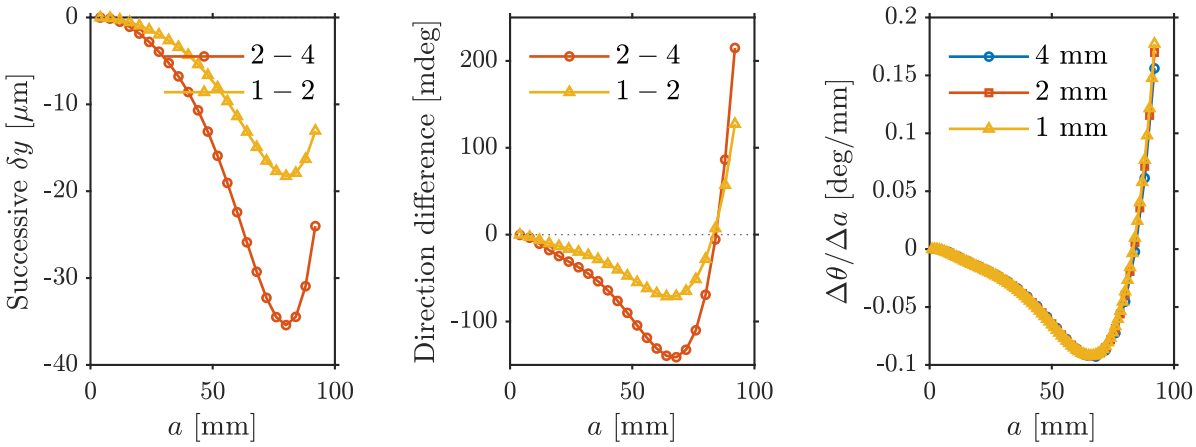
$$q_K = O(\Delta a), \qquad \Delta\theta = O(\Delta a) \tag{25}$$

along a smooth limiting trajectory, while $q_K/\Delta a$ and $\Delta\theta/\Delta a$ remain finite measures of the local corrective response and turning density. Equation (25) is an interpretation of the observed three-level trend, not an independently proved asymptotic result.

The local-symmetry crossing also contracts systematically. The interpolated locations are 84.148913, 83.550120, and 83.268112 mm for the 4-, 2-, and 1-mm paths. The successive shifts



(a) Independently propagated crack trajectories for $\Delta a = 4$, 2, and 1 mm, all continued to 92 mm.



(b) Successive vertical-coordinate differences, $y_{2\,\mathrm{mm}} - y_{4\,\mathrm{mm}}$ and $y_{1\,\mathrm{mm}} - y_{2\,\mathrm{mm}}$, evaluated at exact common crack lengths.

(c) Successive differences in absolute crack direction in the frozen initiation frame.

(d) Turning density $\Delta\theta/\Delta a$ along the three independently propagated trajectories.

Figure 9: Crack-increment sensitivity of the independently propagated LEFM paths. All three calculations use the same dimensionless coarse numerical family, with the absolute core, interaction-domain, transition, and exterior length scales reduced together with Δa. The trajectory panel shows the actual paths, while the two difference panels resolve successive changes at exact common crack lengths. The turning- density panel is plotted on each level's native crack-length grid.

are therefore 0.59879 and 0.28201 mm, whose ratio is 2.12. Interpreted only as a three-level diagnostic this gives an apparent order of about 1.09 for the interpolated event location. The crossing is directly bracketed between the solved 83- and 84-mm states on the 1-mm path, but it remains an interpolated quantity rather than an independently solved zero-K_{II} state.

The combined result is stronger than visual path overlap alone. The trajectory panel establishes the common macroscopic path, while the successive-difference panels show that the remaining geometric and directional discrepancies contract under refinement. At the same time, the raw mode-II correction becomes smaller in proportion to the discrete step. This distinction is important when interpreting q_K: its absolute value along a finite-step MTS path is a step-dependent corrective quantity, whereas its increment-normalized form has the more stable continuous-path interpretation.

6 Discussion

6.1 Interpretation of late redirection

The accepted sequence shows reorientation under increasing K_I. The change in turning sign is directly associated with the change in K_{II}/K_I, as required by Eqs. (10)–(13). The shrinking right-side ligament makes increased free-boundary influence a consistent physical interpretation. However, the boundary distance and the accumulated crack geometry change together. A control with a more distant boundary, or matched fixed-geometry calculations, is needed to isolate that influence. The result supports a mode-mixity explanation of the computed redirection; it does not prove that the boundary is its sole cause.

6.2 Discretization, increment size, and boundary proximity

The 4-mm reference trajectory samples a continuously changing boundary-value problem at discrete geometries, so the finite segment length is part of the numerical propagation rule. The three-level increment study now provides a direct measure of this effect. Successive halving from 4 to 2 to 1 mm reduces the maximum corresponding-tip separation from 35.47 to 18.29 μm and the maximum direction difference from 214.93 to 127.39 millidegrees. The local-symmetry estimate shifts by 0.599 mm and then 0.282 mm. These contractions support a stable limiting trajectory, although three levels are insufficient to establish a formal asymptotic convergence order. The result complements the two-increment comparison of Fang et al. [11], who observed similar but smoother crack paths at smaller prescribed increments: here the effect is quantified over three independently propagated levels and separated into geometric, directional, and mode-mixity measures.

The mode-mixity history clarifies what converges. The raw positive maximum of q_K approximately halves with Δa, whereas $q_{K,\max}/\Delta a$ changes by only about 0.71% over the 4-to-1-mm range; the turning density at the same state is equally stable. This is consistent with the small-mixity MTS relation $\Delta\theta \simeq -2q_K$ and with a smooth limiting path whose curvature is sampled over a finite step. Thus raw q_K should not be interpreted as an increment-independent physical observable of the path. The more robust quantities are the geometry, characteristic event locations, and increment-normalized corrective measures.

The centered-hole control provides a separate check on the directional logic itself. Exact reflection symmetry gives a nearly pure mode-I fixed path, while the measured multiplier $F'(0) = -0.11415$ shows that the implemented recurrence damps small angular perturbations rather than amplifying a numerical mode-II imbalance. This local stability result is consistent with, but logically distinct from, the global increment- refinement evidence for the asymmetric trajectory.

Mesh sensitivity is addressed by two complementary perturbations. The M1 calculation is a clean exterior-only change: it retains the paired tip core and EDI support while substantially

Table 6: Independent-trajectory sensitivity to two mesh changes. The reference trajectory is the production h_0 calculation with the reference exterior mesh. The local-symmetry differences are visualization-only linear interpolation diagnostics between P_{21} and P_{22}.

Perturbation	$\max \lvert\delta y\rvert$ [μm]	$\max \delta r$ [μm]	$\max \lvert\delta\theta\rvert$ [mdeg]	$\max \lvert\delta q_K\rvert$	δa_{LS} [μm]
Coarser exterior M1	0.06734	0.06742	0.07304	2.89×10^{-7}	-0.322
Isolated-core $2h_0$	0.12760	0.12800	0.24539	2.65×10^{-7}	-0.094

reducing exterior resolution, and its independently propagated path remains within 0.0674 μm of the reference tips. The $2h_0$–h_0–$h_0/2$ comparison changes structured-tip resolution with $\eta_e = 1$, holding the requested exterior sizing law fixed while allowing conforming exterior remeshing. The independently propagated $2h_0$ path remains within 0.128 μm of the reference path and preserves the mode-mixity maximum and sign-change bracket.

Table 6 summarizes these two independent mesh-family tests. Their submicrometer path differences are much smaller than the finite-increment differences quantified above, indicating that crack-increment error is the dominant tested numerical sensitivity at the scale of the reported trajectory.

The remaining numerical questions are therefore narrower. Controlled interaction-domain-radius variation is still needed to quantify extraction-domain sensitivity. Likewise, local reflection pairing, synthetic recovery, residual gates, and element-quality gates establish numerical acceptance but do not by themselves bound every physical discretization error. The historical 8-mm verification remains a separate extraction test and is not used as a point in the trajectory-convergence sequence.

6.3 Last accepted state and model scope

The geometry ending at P_{24}, with total polyline length 96 mm, passes geometric and prescribed-field qualification. It has no accepted physical record. The reported attempted PCG solution did not satisfy the strict acceptance criterion; the available compact archive does not retain a failure trace that would quantify its stagnation. The physical history therefore ends at P_{23}, without retrospectively relaxing the gate. This is a numerical limitation, not evidence of fracture arrest or boundary intersection.

The model is quasi-static and linearly elastic. Crack advances are imposed for direction selection; no fracture-toughness growth law, dynamic instability, branching rule, plastic-zone assessment, or subsequent multicrack initiation calculation is provided. These restrictions define the scope of the present trajectory rather than mechanisms excluded by the results.

7 Conclusions

A stress-selected circular-hole point and one prescribed normal segment provide the initial state for an incremental LEFM trajectory whose later directions are determined by interaction-integral SIFs at the current tip and the MTS rule. The 4-mm reference history reaches 92 mm. Positive mode mixity peaks near the middle of the path, later crosses zero, and becomes negative while K_I continues to increase. The sign change reverses the incremental turn and produces the observed late reduction of the negative crack inclination.

The numerical evidence separates several aspects of this recursive path construction. A centered-hole control shows that the exact straight path is a locally attracting fixed path of the implemented EDI–MTS recurrence, with one-step multiplier -0.11415. An independently propagated exterior- mesh perturbation preserves the characteristic mode-mixity events and differs from the reference path by at most 0.0674 μm at corresponding tips. A separate structured-tip

resolution study likewise preserves the mode-mixity maximum and sign reversal; its independent $2h_0$ path differs by at most 0.128 μm. Here $\eta_e = 1$ fixes the requested exterior sizing law while structured-tip resolution changes; conforming exterior connectivity is not required to remain identical.

The crack-increment study provides the strongest trajectory-convergence evidence. Independent 4-, 2-, and 1-mm paths were generated to the same 92-mm length using one dimensionless coarse numerical family. Successive halving of the increment reduces the maximum corresponding-tip separation from 35.47 to 18.29 μm and the maximum direction difference from 214.93 to 127.39 millidegrees. The interpolated local-symmetry location moves from 84.149 to 83.550 to 83.268 mm, with successive shifts of 0.599 and 0.282 mm. At the same time, the peak raw mode mixity decreases almost linearly with the increment, whereas $q_{K,\max}/\Delta a$ and the associated turning density vary by less than 1% across the 4-to-1-mm refinement. The data are therefore consistent with a stable continuous-path limit in which the residual mode-II correction and discrete MTS turn are $O(\Delta a)$ quantities.

These findings support interpretation of the computed trajectory as a robust consequence of the evolving elastic field rather than a particular mesh or finite-step artifact within the tested ranges. The conclusions remain limited to quasi-static LEFM direction selection. The qualified unsolved extension is kept separate from the accepted physical history, and no claims are made about arrest, branching, plasticity, dynamic instability, or subsequent competing crack initiation. Further numerical work should examine interaction-domain-radius sensitivity; a controlled boundary-distance study would be needed to isolate the physical cause of the late redirection.

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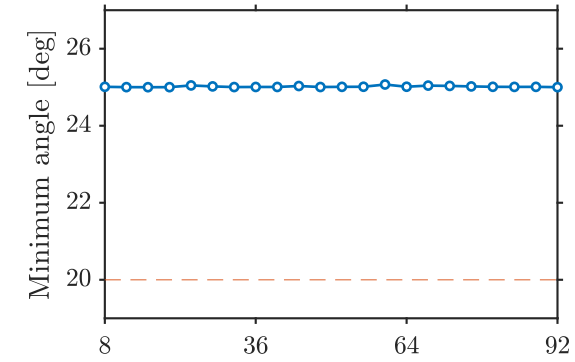
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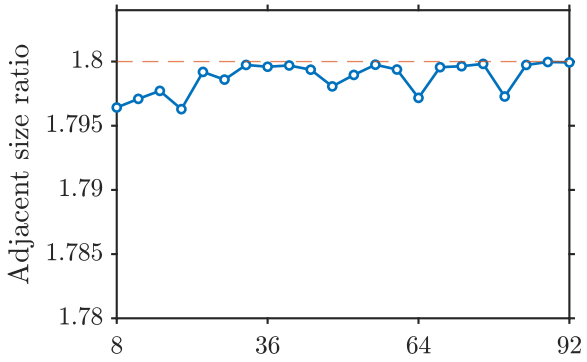
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A Supplementary numerical diagnostics

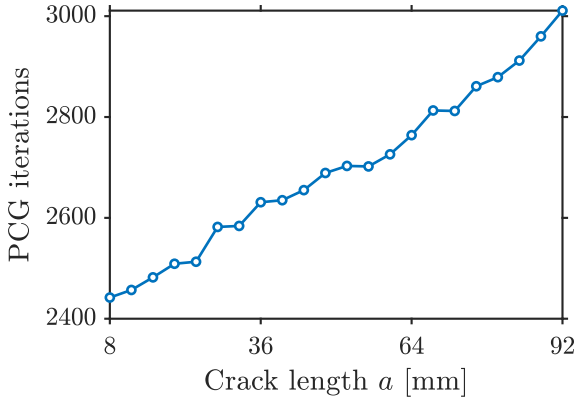
The accepted solver and mesh checks are summarized in Fig. S1. They provide reproducibility and acceptance diagnostics, rather than an estimate of the discretization error in the trajectory.



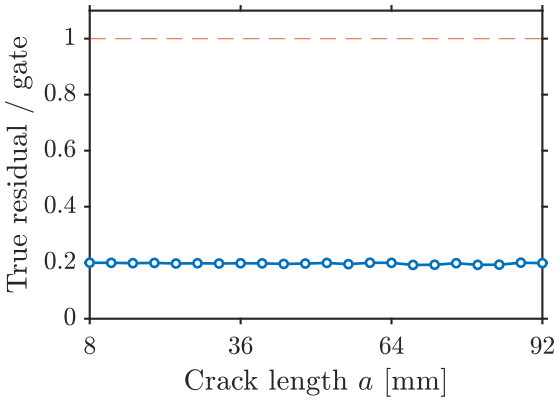
(a) Minimum triangle-angle check.



(b) Adjacent element-size-ratio check.



(c) PCG iterations for accepted physical solves.



(d) True relative residual normalized by its acceptance gate.

Figure S1: Diagnostics for accepted P_2–P_{23} states. Dashed lines are the 20° minimum-angle gate, 1.8 adjacent-size-ratio gate, and normalized true-residual gate of one. The separately enforced reported-PCG tolerance is 10^{-10}; the true-residual acceptance denominator is 5×10^{-10}. No failed P_{24} solve is plotted.